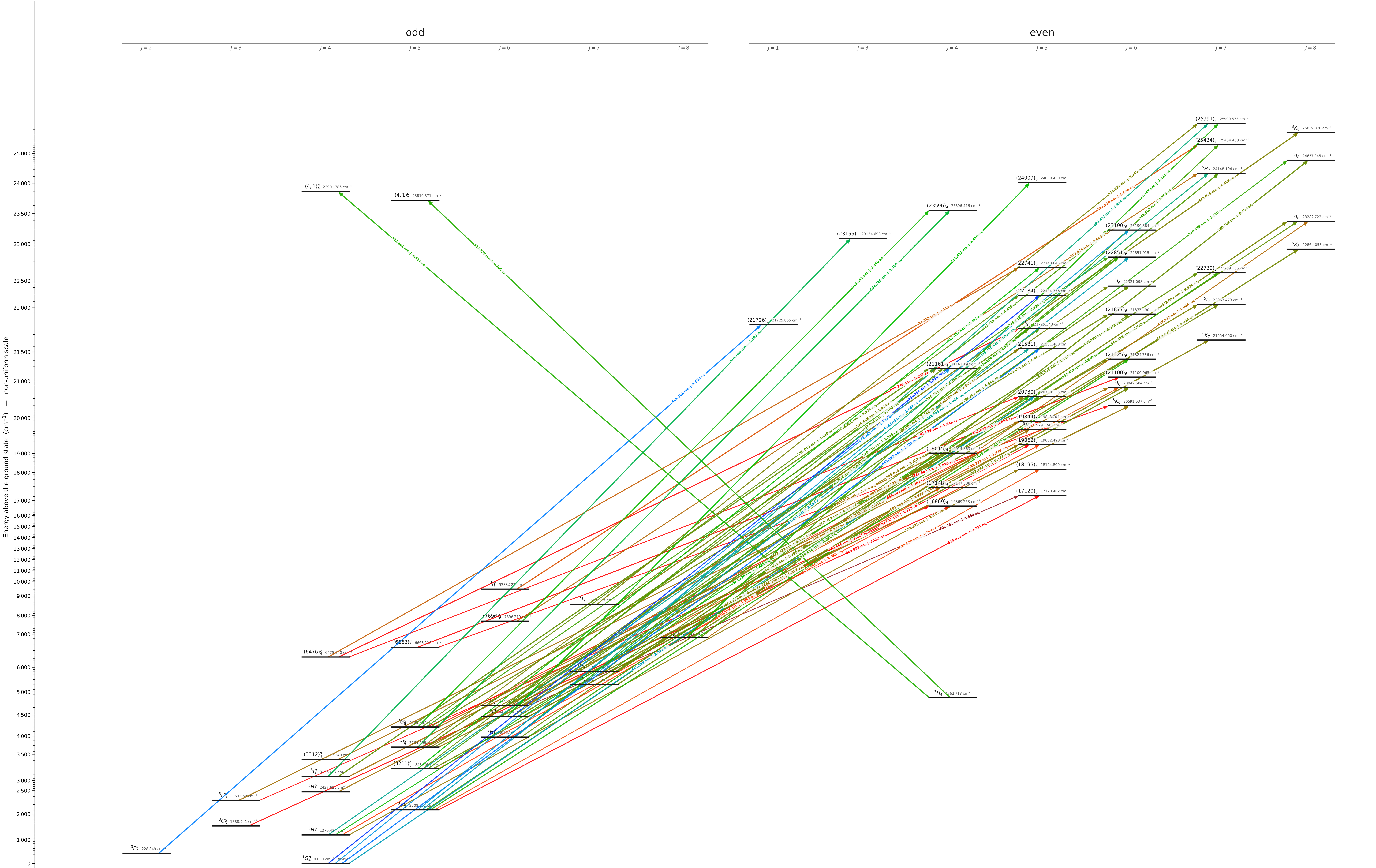
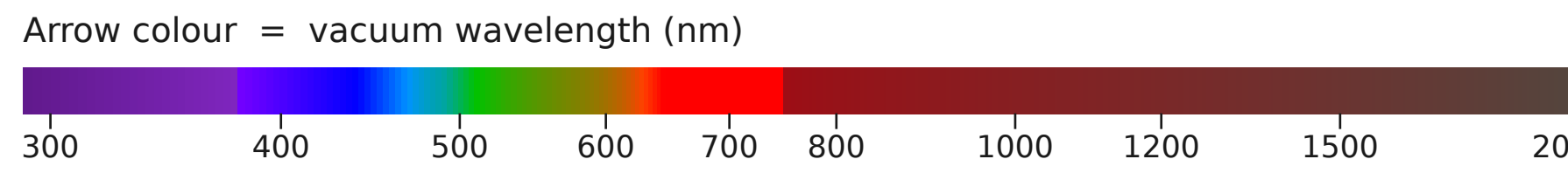


Cerium

Ce (Ce II). Strongest classified transitions below 2 μm (NIST): 62 levels, 77 lines.

Ce + ionisation limit 44672.00 cm⁻¹ = 5.53862 eV (223.854 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory; ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
³ f _g - (²³¹⁵⁵) ₃	501.0501	500.9103	598.3284	5.191	6.200e+07	-	-	-
³ f _g - (²³⁵⁹⁶) ₄	504.2252	504.0846	594.5606	5.000	4.390e+07	-	-	-
³ f _g - (²⁴⁰⁰⁹) ₅	511.4128	511.2704	586.2044	4.976	3.410e+07	-	-	-
³ f _g - (²⁵⁹⁹¹) ₇	521.3372	521.1920	575.0453	7.111	4.820e+07	-	-	-
³ H ₄ - (⁴ ,1)g ²	522.4915	522.3460	573.7748	6.417	6.500e+07	-	-	-
³ H ₄ - (⁴ ,1)g ²	524.7374	524.5913	571.3191	6.206	4.910e+07	-	-	-
³ Hf ₅ - ³ H ₇	545.0754	544.9239	550.0018	6.403	3.420e+07	-	-	-
³ f _g - (²¹¹⁶¹) ₄	556.6508	556.4962	538.5647	5.625	4.130e+07	-	-	-
³ f _g - ³ I ₅	556.7513	556.5968	538.4674	5.076	2.750e+07	-	-	-
³ f _g - ³ I ₆	560.2832	560.1277	535.0731	9.794	6.500e+07	-	-	-
(³²¹¹)g ² - ³ I ₆	567.1532	566.9958	528.5917	6.373	3.470e+07	-	-	-
³ f _g - ³ K ₆	569.8573	569.6993	526.0834	8.934	5.140e+07	-	-	-
³ Hf ₅ - ³ I ₆	572.0623	571.9036	524.0556	8.924	5.070e+07	-	-	-
³ f _g - ³ K ₆	578.9750	578.8145	517.7986	8.426	4.360e+07	-	-	-
³ f _g - ³ K ₇	581.4527	581.2915	515.5922	6.608	3.000e+07	-	-	-
³ f _g - ³ K ₆	594.2502	594.0856	504.4886	6.107	2.770e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).